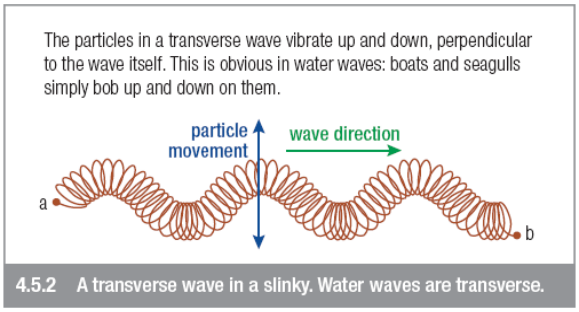
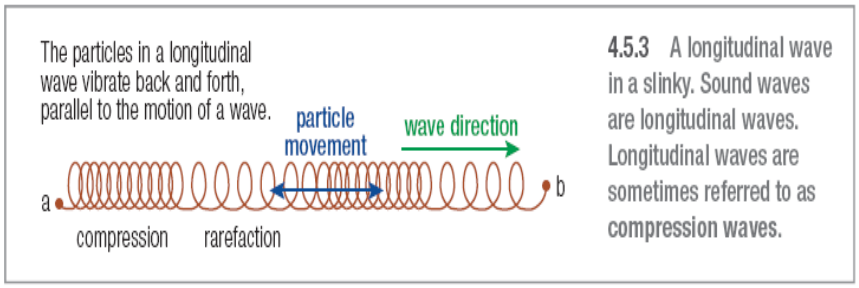
ACTIVITY 1: Use the PowerPoint presentation to complete the blank spaces below...

Light as a Wave

Light travels in _____ lines at a _____ of just below _____ kilometres per second. _____ else can reach this _____ or travel faster than the speed of _____. Unlike sound, _____ does not require particles (such as those found in _____, air or gas) to _____ through. This means that light can pass _____ the vacuum of _____ while sound cannot. Light can do this because it transmits its _____ as a special type of _____—an electromagnetic wave. Two kinds of waves - classified as either _____ or _____.



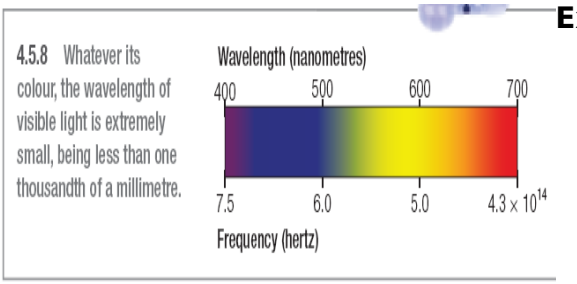
4.5.2 A transverse wave in a slinky. Water waves are transverse.



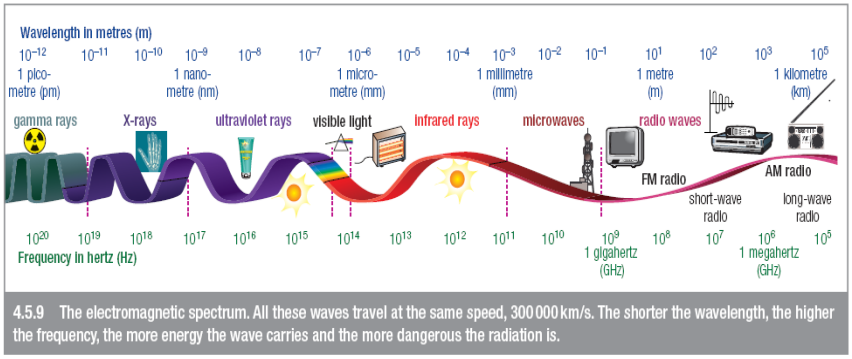
4.5.3 A longitudinal wave in a slinky. Sound waves are longitudinal waves. Longitudinal waves are sometimes referred to as compression waves.

Visible light

The **visible** _____is the **r**ed, **o**range, **y**ellow, **g**reen, **b**lue, **i**ndigo and **v**iolet (_____) seen in the _____. Each colour has a different _____ and a different frequency. Special cells called _____ in the _____ of the eyes respond to _____ wavelengths and determine which _____ are seen.



Explosions in space



4.5.9 The electromagnetic spectrum. All these waves travel at the same speed, 300 000 km/s. The shorter the wavelength, the higher the frequency, the more energy the wave carries and the more dangerous the radiation is.

In sci-fi movies, _____ in space are usually _____ shown as being bright and _____. Although _____ can pass through a vacuum, _____ cannot. Explosions in space are _____ and visible but completely _____.

Jam session beneath the sea

_____ seems to be a very _____ place to humans but this is because our _____ are normally full of _____. Divers often hear constant _____when in the sea. Although it _____ like bacon cooking, it is _____ the sound of shrimp cracking their _____, forming a small bubble that then '_____'. Scientists from Curtin University of WA have _____ thousands of underwater _____. Shrimp, fish and sea urchins form loud _____. Some sound like electric guitars, while others sound like _____, monkeys, lightning storms or a lion's _____. It's noisy because _____ travels further in _____ than it does in _____.